

The Financial Opportunity Cost of Continued Navy Service

A General Internist Case

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Individual Research Paper - a global or local challenge, analyzed

Challenge and approach

The Navy needs to retain physicians without assuming that each civilian salary advantage requires an equivalent bonus. Across DoD medical facilities, assigned military medical personnel fell from 60,479 in FY2015 to 50,590 in FY2023; Naval Medical Center Portsmouth fell from three general pediatricians to one in summer 2023 (U.S. Government Accountability Office [GAO], 2025). These figures describe staffing pressure, not Navy-wide physician departures. GAO (2020) found military cash pay below civilian medians in 21 of 27 medical and dental specialties. Its tracker reports a June 2026 DOD update; recommendations on replacement costs, retention data, and civilian wages remain open.

Opportunity cost is what the physician gives up by staying. Past medical-school support is a sunk cost; decisions at the margin compare the benefits and costs of additional service. Compensating wage differentials describe pay differences associated with job amenities and burdens (Frank et al., 2019). This case compares an O-4 general internist leaving at 10 active-service years with remaining to 20. The physician is married, age 37, and stationed in San Diego. Marit Health's (2026) \$459,057 local compensation average is a benchmark, not a matched offer. The model treats it as civilian employee cash pay and adds benefits separately. Appendix A specifies all inputs.

Both comparisons use the same O-4 longevity increases and pension base. Present value converts future amounts to today's dollars: cash flows are discounted at 5% real and pension payments at an illustrative 3%. The latter reflects inflation protection, but alternative rates are tested. All amounts are in constant 2026 dollars; reaching retirement eligibility is assumed, not guaranteed.

Results and their meaning

The four-year gross-pay comparison produces a \$636,209 present-value civilian advantage, but stops before pension eligibility. Across the full ten years, the gross civilian advantage is \$1,349,196, offset by a conditional pension worth \$733,228. Thus, staying costs \$615,968 on the gross basis. Figure 1 shows this cost falling as retirement approaches, with break-even near 14.3 years. These calculations include longevity increases without promotion; Appendix B makes the annual arithmetic visible.

Taxes and benefits substantially change that comparison. Tax-free housing and subsistence allowances favor Navy pay, while civilian wages face income and payroll taxes. After adding employer retirement contributions and subtracting civilian insurance and health premiums, staying costs \$58,419. Break-even moves to 10.9 years. This is a tax-adjusted compensation comparison, not spendable income: retirement contributions are included at face value, and later withdrawal taxes are omitted. Appendix C identifies the tax brackets and simplifications.

Figure 2 then adds retiree health coverage, bringing staying to approximately break-even. A qualifying Navy domicile without state income tax raises its value to \$98,456; assumed continuation pay raises it to \$114,808. A 5% civilian earnings penalty after military retirement lowers it to \$29,072. These are conditional scenarios, not a confidence interval. The near-zero results should not be presented as proof that staying is financially superior.

Policy implications and existing services

The gross-adjusted difference motivates testing understanding, not assuming a bonus effect. The Medical Corps already expects O-3 through O-5 physicians, excluding interns, to complete a Career Development Board (CDB) each tour (Schofer, 2025). Navy financial counseling, Beyond Basic Pay, and special-pay guidance provide additional resources (MyNavy HR, n.d.-a; Department of Defense, n.d.-b; Navy Medicine, n.d.). The proposed addition is a standardized financial review tied to obligation completion and evaluated for retention.

Schofer (2021) illustrates pension value when weighing continued service; this supports the topic, not a retention effect or endorsement of this pilot. NHS Employers (2022, 2025) supplies adaptable statements and communication guides. At United Lincolnshire Hospitals, 18 retiring staff returned during a broader retention effort; education's effect cannot be isolated (NHS Employers, 2020). Randomized military and university experiments show effects on retirement-plan participation, not physician retention (Goldin et al., 2020; Duflo & Saez, 2003). Financial education also has limits. DODIG reports that a January 2024 BUMED emergency-physician study identified concerns about skill degradation as the "number-one factor contributing to junior officer attrition" (Department of Defense Office of Inspector General, 2025, p. 15). Such findings inform questions about departures but cannot establish a counseling effect or be generalized directly to internists.

The model is sensitive: a 5% pension discount raises the adjusted cost of staying to \$234,384; an added 9.3-percentage-point pension state tax raises it to \$126,609. Promotion, deductions, retirement-account taxation, separation risk, and unmatched salary benchmarks remain limitations. This model estimates financial opportunity cost; it does not assign monetary values to job satisfaction, professional identity, family preferences, deployment stress, or commitment to service. These factors may outweigh the financial difference and should be considered during counseling.

Recommendation and decision rule

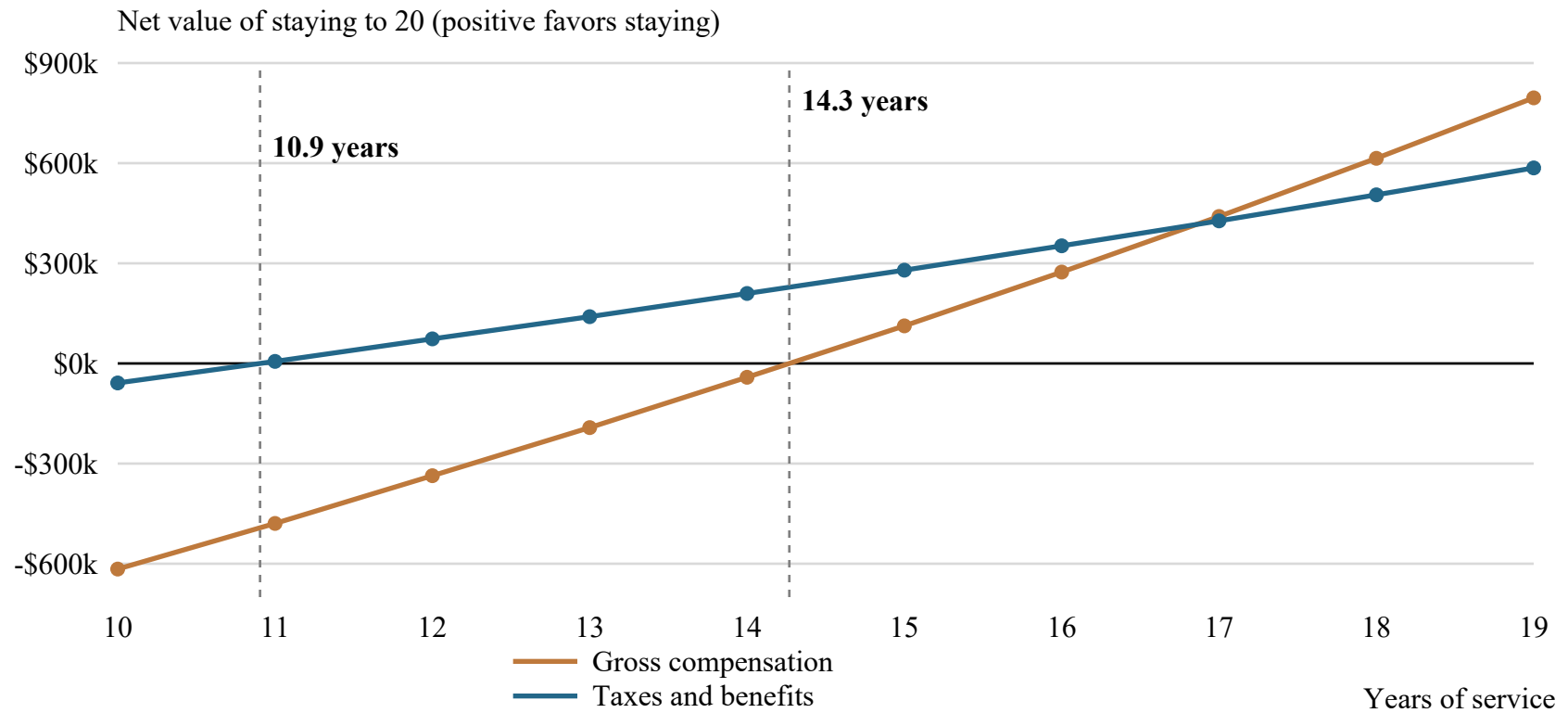
Pilot a standardized compensation review within the Medical Corps CDB program for physicians whose obligations end within 12 months. Require enhanced-group commands to document an offer before extension or separation commitment. The CDB coordinator arranges access; the Command Financial Specialist provides education and referral; a financial counselor reviews the comparison; personnel specialists verify eligibility and obligations. Keep household finances and civilian offers private, with voluntary disclosure.

First establish whether eligible physicians materially misunderstand benefits using the Appendix F knowledge questions and interviews. Test comprehension, uptake, delivery time, and recruitment before a retention trial. Appendix F gives illustrative sample and cost calculations, not evidence of available participants or approved funding. If misunderstanding is uncommon, or recruitment is inadequate, reassess the intervention rather than claiming a retention test is feasible.

If feasible, randomize commands to existing services or those services plus the review. Balance specialty and service stage; analyze by assigned group with command clustering. Preserve usual services and record their use. Measure knowledge at baseline and six months, and active-service status 24 months after the original obligation ends.

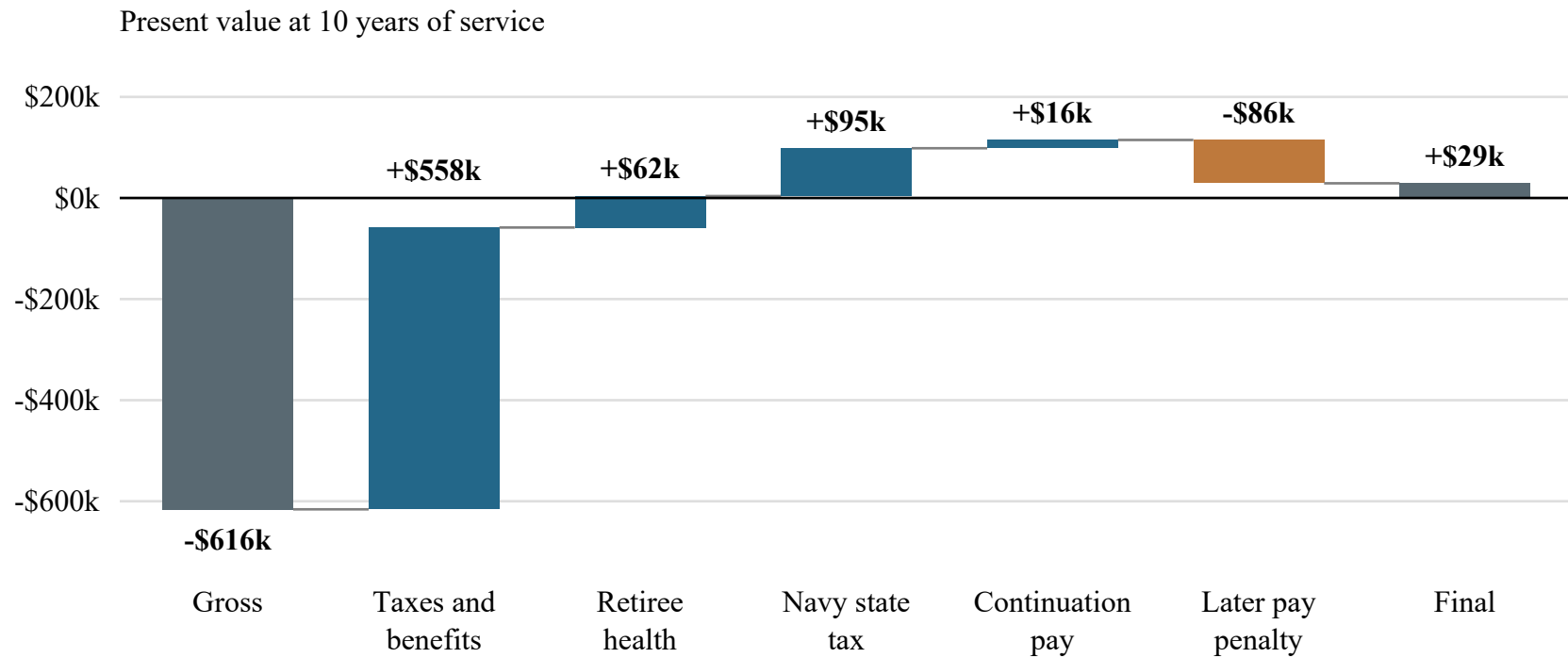
Before enrollment, approve expansion criteria: a five-percentage-point retention improvement, a 95% confidence interval excluding zero, and cost per additional physician retained below 25% of documented specialty replacement cost. These are policy judgments. Count delivery, verification, updates, administration, and evaluation costs. If knowledge improves without retention, reconsider pay and working conditions. Report inconclusive effects without claiming success.

Figure 1 Break-even depends on what is counted



The same O-4 longevity pay and \$50,274 annual pension are used on both paths. Pension: 30 payments discounted at 3%; pre-retirement compensation: 5%. The adjusted line excludes retiree health, continuation pay, and later civilian pay differences. Zero crossings are interpolated. Near year 17, gross value overtakes adjusted value because the remaining pay gap shrinks while the gross pension remains untaxed. Neither crossing estimates a retention effect. Calculations: Appendices A-D.

Figure 2 Taxes and benefits narrow the difference



Each step adds to the preceding result. Positive values favor staying. Navy state tax means a qualifying domicile without state income tax during active duty. Later pay penalty means 5% lower civilian pay after military retirement, through age 65. Values are rounded; exact totals and assumptions appear in Appendix D.

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Appendix A Compensation and assumptions

Table A1. O-4 annual cash compensation by completed service years. BAH is \$5,082 per month for ZIP 92134 with dependents. BAS is \$328.48 per month. Incentive pay, board certification pay, and retention bonus total \$99,000 per year.

Completed years	Basic pay	Navy gross cash
10-11	\$113,040	\$276,966
12-13	\$118,660	\$282,585
14-15	\$122,573	\$286,499
16-17	\$124,819	\$288,745
18-19	\$126,119	\$290,045

Sources: DFAS (2026a-c), DTMO (n.d.). Pay rates are held in 2026 dollars. The \$48,000 annual retention bonus assumes continued eligibility and agreements covering the full comparison horizon; it is not guaranteed. No promotion is assumed.

Civilian employee cash pay is \$459,057 each year (Marit Health, 2026). Current age is 37; retirement occurs at 47; work ends at 65. Married filing jointly, one income, no child credits. No VA benefits, survivor coverage, pay growth, transition cost, or first-year pay reduction is included.

Appendix A Benefit assumptions continued

Item	Annual amount or assumption
BRS service contribution	5% of basic pay; full eligibility and employee contributions assumed
Civilian employer match	\$16,560 = 4.6% of min(\$459,057, \$360,000)
Civilian worker health premium	\$6,850 family premium share
Private disability insurance	\$13,771.71 = 3% of civilian pay
Retiree health sensitivity	\$6,085 = \$6,850 avoided premium less \$765 fee
Continuation pay sensitivity	\$23,550 in comparison year 2; 2.5 months of basic pay

Sources: DoD (2017), Vanguard (2025), IRS (2025b), KFF (2025), TRICARE (2026), and The White Coat Investor (n.d.). Continuation pay is a conditional scenario assumption; eligibility, future rates, and obligation terms must be confirmed.

Employer contributions are compensation value, not current cash. Their future withdrawal taxes and employee contribution deductions are excluded. Insurance at 3% is an illustrative quote proxy. Health savings ignore deductibles, copays, and differences in access. These assumptions require individual verification.

Appendix B Annual adjusted comparison

Table B1. Dollars rounded for display; calculations use unrounded values. Civilian adjusted compensation is \$305,005.33 every year. Navy tax includes federal, California, and employee payroll taxes. Navy value equals gross cash minus taxes plus the BRS service contribution.

Year	Navy gross	Navy tax	Navy value	Civilian gap
1	276,966	55,069	227,549	77,457
2	276,966	55,069	227,549	77,457
3	282,585	56,909	231,609	73,396
4	282,585	56,909	231,609	73,396
5	286,499	58,191	234,436	70,569
6	286,499	58,191	234,436	70,569
7	288,745	58,927	236,059	68,946
8	288,745	58,927	236,059	68,946
9	290,045	59,352	236,998	68,007
10	290,045	59,352	236,998	68,007

The present value of the annual civilian gaps at 5% is \$557,014.21. The corresponding gross gaps total \$1,349,196.04 in present value. For the first four years, the gross-gap present value is \$636,208.84. All payments occur at year end. Tax and civilian benefit components are specified in Appendix C.

Appendix C Tax calculation inputs

Federal income tax uses 2026 married-filing-jointly brackets and a \$32,200 standard deduction (IRS, 2025a). California uses the published 2025 Schedule Y and \$11,080 standard deduction as a proxy held constant; these are not verified 2026 California brackets (FTB, 2025a, 2025b).

Marginal rate	Federal lower bound	California lower bound
10% federal / 1% CA	\$0	\$0
12% / 2%	\$24,800	\$22,158
22% / 4%	\$100,800	\$52,528
24% / 6%	\$211,400	\$82,904
32% / 8%	\$403,550	\$115,084
35% / 9.3%	\$512,450	\$145,448
37% federal	\$768,700	Not used

Apply each marginal rate only to taxable income within its bracket. California income remains below the next \$742,958 threshold in all reported cases. State exemption credits, itemization, SALT interactions, and alternative minimum tax are excluded. The model is an illustration, not a tax return.

Appendix C Worked example and payroll taxes

Year 1 Navy taxable wages are \$276,965.76 minus \$64,925.76 in BAH and BAS, or \$212,040 (IRS, 2025c). Civilian wages are \$459,057. Taxable income for each tax is wages minus that tax system's standard deduction.

Year 1 component	Navy	Civilian
Federal income tax	\$28,989	\$89,506
California income tax	\$11,567	\$34,539
Employee payroll tax	\$14,514	\$25,945
Total modeled taxes	\$55,069	\$149,990

Payroll tax = 6.2% of wages up to \$184,500, plus 1.45% of all wages, plus 0.9% above \$250,000 for this joint household. California civilian wages also incur 1.3% SDI (SSA, 2026; EDD, 2026).

Military allowances are excluded from wages.

Civilian value = \$459,057 - \$149,989.96 + \$16,560 - \$6,850 - \$13,771.71 = \$305,005.33. Navy year-1 value = \$276,965.76 - \$55,068.94 + \$5,652 = \$227,548.82. Difference = \$77,456.51.

Appendix D Pension and present value

Define $A(r,n) = [1 - (1+r)^{-n}] / r$. For n remaining years, the pension value is annual pension times $A(0.03,30)$, divided by $(1.03)^n$. Each remaining annual compensation gap is divided by $(1.05)^t$, with t starting at 1.

High-3 basic pay = $(\$124,819.20 + \$126,118.80 + \$126,118.80) / 3 = \$125,685.60$. The BRS annual pension is 2% times 20 years times High-3 = $\$50,274.24$ (DoD, n.d.-a). The first of 30 pension payments occurs one year after retirement.

At 10 years, gross pension PV = $\$733,228.13$. Subtracting the gross remaining pay gap of $\$1,349,196.04$ gives $-\$615,967.91$. Applying a flat illustrative 32% federal pension tax gives pension PV of $\$498,595.13$. Subtracting the adjusted compensation gap of $\$557,014.21$ gives $-\$58,419.08$.

At each later service year, discard completed compensation years and discount the remaining gaps from that decision date. Keep the same retirement pension. Interpolate between the last negative and first positive net value. This gives 14.27 years for gross compensation and 10.90 for the adjusted comparison.

No pension probability adjustment is made. A 32% pension tax for all 30 payments is a simplification: actual rates depend on later earnings, deductions, residence, and tax law.

Appendix E tests a higher combined pension-tax assumption.

Appendix D Cumulative adjustments

Table D1. Net value of staying at 10 years, rounded to dollars; each row adds the stated change. Positive values favor staying. The cash and pension paths share the same longevity schedule and High-3 base.

Comparison	Net value
Gross compensation; untaxed pension	-\$615,968
Taxes and current benefits	-\$58,419
Add retiree health value	\$3,854
Add no-income-tax Navy domicile	\$98,456
Add assumed continuation pay	\$114,808
Add 5% later civilian pay penalty	\$29,072

Retiree health PV = \$6,085 times $A(0.03,18) / (1.03)^{10} = \$62,273.31$. Removing active-duty Navy state income tax saves \$94,602.25 in PV. Continuation pay contributes \$16,351.50 after incremental federal and payroll tax, discounted two years.

The later earnings penalty costs \$85,736.06 in PV. It compares civilian pay of \$459,057 and \$436,104.15 after modeled wage taxes, employer match, and 3% insurance, for 18 years after retirement. Fixed health premiums cancel. These scenarios are not additive beyond the sequence shown.

Appendix E Sensitivity and model boundaries

Change from adjusted base	Net value of staying
Base: California wages, 3% pension discount	-\$58,419
Pension discount 5%	-\$234,384
Civilian salary \$400,000	\$201,025
Civilian salary \$350,000	\$436,422
Pension tax 41.3% instead of 32%	-\$126,609

Each sensitivity changes one input. Lower civilian salary also changes its taxes and 3% insurance cost. Pension tax of 41.3% adds an illustrative 9.3-percentage-point state burden; it is not a forecast of residence or a complete future tax calculation.

The benchmark combines employer types and experience levels. The model treats it as an employee offer with separately specified benefits; an actual offer may have different hours, call, productivity risk, retirement terms, or insurance coverage. Those differences must be checked before drawing an individual conclusion.

The comparison does not estimate retention elasticity, command vacancies, replacement costs, or readiness outcomes. Financial break-even is not evidence of a bonus effect. Promotion, clinical re-entry, medical separation, future pension taxation, and personal preferences can change the result.

This model estimates financial opportunity cost; it does not assign monetary values to job satisfaction, professional identity, family preferences, deployment stress, or commitment to service. These factors may outweigh the financial difference and should be considered during counseling.

Appendix F Existing services and the pilot gap

Medical Corps Career Development Boards (CDBs) provide the delivery foundation. A December 2025 announcement reiterates the once-per-tour expectation for O-3 through O-5 physicians, excluding interns (Schofer, 2025). The pilot adds a standardized financial review timed to obligation completion and a retention evaluation. That added requirement is proposed, not existing policy.

Financial counseling. Navy milestone training and individual counseling already cover finances; continuation-pay counseling addresses eligibility and obligations. The pilot adds a standardized stay-or-leave comparison (Commander, Navy Installations Command, n.d.; Financial Readiness Program, n.d.; MyNavy HR, n.d.-a).

Compensation information. The Beyond Basic Pay campaign explains total rewards, and Navy Medicine publishes special-pay guidance. The pilot applies these to household taxes, specialty pay, pension timing, and civilian offer terms (Department of Defense, n.d.-b; Navy Medicine, n.d.). DFAS (2021) directs Navy personnel to the RMC calculator; an annual Navy myPay compensation statement is not assumed.

Transition preparation. TAP initial and pre-separation counseling generally begin at least 365 days before release. Early timing already exists; the pilot focuses on evaluating continued service before commitment, while preserving access to TAP (MyNavy HR, n.d.-b).

Retention research. DODIG reports that a January 2024 BUMED emergency-physician study identified concerns about skill degradation as the "number-one factor contributing to junior officer attrition" (Department of Defense Office of Inspector General, 2025, p. 15). Whether it is the survey announced in 2023 is unconfirmed. Existing findings inform questions; the pilot collects its own baseline and comparison outcomes.

Appendix F Adapted counseling materials

F1-F7 are proposed templates, not validated instruments. They adapt NHS Employers (2022, 2025), the Financial Readiness Program (n.d.) checklist, and existing Navy and DoD resources described above.

F1. Personalized compensation statement (Figure F1). Record specialty, service years, obligation end date, retirement system, household assumptions, and source dates. Compare leaving when eligible with staying to retirement over a common horizon. Separate annual after-tax cash, employer contributions, and pension present value. Include allowances, special pays, bonus obligations, insurance, and actual civilian offer terms; label benchmarks. Show sensitivity and permit corrections. This pilot statement is not represented as a Navy myPay product.

F2. Counseling checklist. Verify eligibility and benefit rules with personnel specialists; explain contingent benefits and added obligations; check civilian hours, call, productivity terms, and benefits. Review taxes and pension assumptions; ask the physician to explain the tradeoff; record corrections, unresolved questions, and follow-up. Financial counselors use a common model with specialist support.

F3. Reasons for leaving. Rank compensation, workload, deployment, clinical practice and skill maintenance, autonomy, location, and family needs; allow other reasons. Ask which change could affect the decision. Measure benefits knowledge separately from satisfaction. Historical surveys guide question design, not the participant baseline or control group.

Appendix F CDB delivery and responsibilities

F4. Delivery plan. Enhanced-group commands would be required to schedule and document an offer of review within 12 months before obligation completion and before an extension or separation commitment. An earlier routine CDB would not automatically satisfy this proposed review. Declining private financial disclosure would not affect access to career guidance.

The command Medical Corps CDB coordinator identifies eligible officers, arranges the review, and records the offer, acceptance or decline, and completion. Existing remote CDB arrangements offer a route for officers without a local coordinator (Schofer, 2024). The pilot lead maintains the audience, channel, owner, and date schedule adapted from NHS Employers (2022).

The Command Financial Specialist provides initial compensation education, the checklist, and referral. The CFS instructor guide describes education, counseling, and referral functions (Financial Readiness Program, 2021). A Fleet and Family Support Center financial counselor reviews the individualized model with specialist support; CFS designation alone is not treated as qualification to validate complex physician tax and pension estimates.

Personnel or special-pay specialists confirm eligibility, agreement dates, and added obligations.

The CDB discusses clinical work, deployment, career goals, and family priorities. Detailed household figures and civilian offers remain in the private counseling process; the board receives only information the physician chooses to share.

Record offers, uptake, session duration, model version, unresolved questions, and follow-up.

Train staff on the same materials and budget for their time. These responsibilities are proposed additions to existing services and require pilot authorization.

Appendix F Comparison groups and knowledge

Existing-services commands retain usual Medical Corps CDBs, financial counseling, TAP, special-pay guidance, and general compensation education. Enhanced-group commands retain all of those services and add the standardized review and documented offer. The evaluation estimates the effect of offering this package; it does not withhold existing assistance.

Record existing-service use, outside counseling, shared materials, and review uptake in both groups. Keep physicians in assigned groups even if they decline the review. Do not compare only volunteers with nonparticipants and describe that difference as a randomized treatment effect.

F5. Knowledge measure. At enrollment before the offer and six months later, administer the same five questions in both groups: identify taxable versus tax-exempt pay; identify pension eligibility; explain the bonus obligation; distinguish annual cash from pension present value; and identify an assumption that could reverse the comparison. Score zero to five using a prespecified rubric and current rules.

Test comprehension and scoring consistency before enrollment. Record missing answers and follow-up loss. Collect service stage, specialty, obligation end date, and baseline intention to stay.

A knowledge increase is an intermediate outcome; it does not establish a retention effect.

Appendix F Retention evaluation and costs

F6. Outcome record. Randomize commands, balance specialty and service stage, and analyze physicians in assigned groups with command clustering. Prespecify eligibility, required sample size, exclusions, and missing-data handling. Track participation and knowledge separately from service outcomes.

The primary retention endpoint is active-service status 24 months after the original obligation ends. Record voluntary departures, involuntary losses, and reasons. Compare the offer of enhanced review with contemporaneous existing CDB and financial services; historical exit surveys cannot substitute for a randomized comparison. Document survey provenance before using any existing item or result.

F7. Cost and decision record. Count incremental CDB coordination, CFS and counselor time, physician-pay verification, model development and updates, administration, data collection, and evaluation. Record setup and recurring costs separately. Existing resources may support delivery; cost savings and cost-effectiveness must be measured.

Divide incremental cost by estimated additional physicians retained at the endpoint only if the estimated effect is positive. Report uncertainty; do not claim cost-effectiveness when retention effects are indistinguishable from zero. Prespecify the main-text five-percentage-point, confidence-interval, and replacement-cost criteria before results. These proposed policy thresholds require approval and are not validated by the source programs.

Interpret improved knowledge without retention as evidence to reassess the intervention and other departure drivers, including clinical skill maintenance and family needs. Report adverse, null, and inconclusive findings as well as favorable results.

Appendix F Illustrative counseling handout

Figure F1 Sample compensation statement

Illustrative counseling handout—not a validated instrument

O-4 internist | 10 service years | Age 37 | Married, one income

Decision: leave when eligible or remain to 20 service years

Year 1 annual compensation	Stay	Leave
Gross cash pay	\$276,966	\$459,057
After taxes and listed premiums	\$221,897	\$288,445
Employer retirement contributions	\$5,652	\$16,560
Adjusted compensation total	\$227,549	\$305,005

Cash excludes employee retirement saving and other spending.

Retirement value and full-horizon results

Conditional military pension PV if staying: \$498,595 after tax

Adjusted base case: leaving is ahead by \$58,419 in present value.

Retiree health: +\$3,854; plus Navy tax domicile: +\$98,456.

With continuation pay: +\$114,808; later pay penalty: +\$29,072.

Positive favors staying; cumulative scenarios, not a confidence interval.

Assumptions: CA wages; pension at 20; recurring bonus eligibility.

Discounts: cash 5%, pension 3%; pension tax 32%; no VA benefits.

5% pension discount: -\$234,384. Check offer terms and family needs.

Confirm obligations, eligibility, domicile, and clinical practice needs.

Note. Rounded base-case values from Appendices A-E. The pension value is separate from annual compensation. Layout adapted from NHS Employers (2022, 2025); this example is not an individual recommendation.

Appendix F Benchmark and feasibility gate

F8. Benchmark and inference checks. The Marit Health (2026) page, checked October 8, displays the \$459,057 all-employer average alongside base pay of \$445,914 and bonuses of \$13,142 (a \$1 display-rounding difference). That supports a cash-pay reading, but does not establish an employee-only matched offer. The visible examples include both W-2 and 1099 arrangements. The benchmark is therefore retained as an illustrative input, not verified individual compensation.

Before counseling, record a written offer with cash salary, expected bonus, hours, call, employment status, employer retirement contributions, health premiums, disability and malpractice coverage, and source date. Add only benefits excluded from the quoted cash amount; do not apply employee payroll-tax rules to a contractor offer. The \$400,000 and \$350,000 rows in Appendix E remain hypothetical salary sensitivities, not newly verified offers.

The robust conclusion is that taxes and benefits substantially narrow the gross compensation gap under the stated assumptions. The sign of the near-zero result is not robust. Retiree health produces +\$3,854 in the cumulative sequence, while a 5% pension discount produces -\$234,384 from the adjusted base. These are different scenarios, not uncertainty bounds or probabilities. No matched-offer or retention-effect claim follows from either.

F9. Feasibility gate. Invite an illustrative 60 eligible physicians across six commands to test the five-question instrument, interview misunderstandings, and measure uptake and delivery costs. This is a planning cohort, not a powered retention trial or a representative prevalence estimate. Use results to prespecify what constitutes a consequential knowledge deficit. Proceed only if such a deficit, acceptable delivery, and a sufficient eligible command population are documented. A small cohort cannot establish a five-point retention effect.

Appendix F Trial planning assumptions

F10. Illustrative retention-trial size. Assume control retention of 50%, enhanced retention of 55%, equal allocation, two-sided alpha 0.05, and 80% power. The normal approximation gives 1,564.7 evaluable physicians per arm before clustering. With mean enrolled cluster size $m = 20$ and intraclass correlation $\rho = 0.02$, the design effect is $1 + (m - 1)\rho = 1.38$. Allow 10% outcome loss and round upward to whole commands: 120 commands per arm, 240 total, and 4,800 physicians enrolled.

Calculation: $n \text{ per arm} = [1.960 \sqrt{2 \bar{p}(1-\bar{p})} + 0.842 \sqrt{p_0(1-p_0)+p_1(1-p_1)}]^2 / (p_1-p_0)^2$, with $\bar{p} = (p_0+p_1)/2$. Commands per arm = $\text{ceiling}[n \times \text{design effect} / (0.90 \times 20)]$. Using ρ of 0, 0.01, and 0.05 gives 174, 208, and 340 commands respectively. All rates, cluster sizes, and correlations are planning assumptions, not measured Navy inputs.

This scale may exceed the available eligible population; no command or physician inventory has been established. Do not promise a definitive trial. Obtain actual counts and baseline rates, then have a statistician model unequal clusters, finite-command inference, attrition, and any stratification before authorization. Detecting a five-point effect against zero also does not guarantee meeting a rule requiring the observed effect itself to be at least five points.

F11. Illustrative costs. Assume \$20,000 setup: 80 hours for model preparation, 40 for training, and 80 for evaluation design, each at \$100/hour. Add \$1,200 participant assessment time (60 x \$20), \$3,000 interviews and instrument testing, and \$6,000 statistical planning. These are invented planning rates and allowances, not Navy salary estimates or vendor quotes.

Appendix F Illustrative budget and limits

For each enhanced offer, budget coordinator time (0.5 hour x \$60), initial education (0.5 x \$75), counselor review (1.5 x \$100), specialist verification (0.5 x \$100), and follow-up (0.5 x \$60): \$297.50. Budget another \$30 per enrolled physician for outcome collection in either group. The six-command, 60-person feasibility illustration, with 30 enhanced offers, totals \$40,925. It assumes full uptake for costing and includes participant assessment time separately.

For the illustrative 4,800-person trial, applying the same allowances yields \$888,200: \$30,200 setup and planning, \$144,000 outcome collection, and \$714,000 for 2,400 enhanced offers. With 10% outcome loss and a true five-point effect, the enhanced arm would have an expected 108 additional retained physicians among observed outcomes. The illustrative cost is \$8,224 per additional physician; the proposed 25% ceiling would require documented replacement cost above approximately \$32,896.

These are conditional planning ratios, not measured cost-effectiveness. Count total incremental evaluation costs in both arms and additional retention in the enhanced arm; do not multiply the effect by both arms. Lost outcomes require prespecified handling, not an assumption that missing physicians left service. Recalculate with actual staffing rates, travel, overhead, multi-year follow-up, and model-maintenance costs. Never use the small feasibility cohort to claim a retention return on investment.

Source and reproducibility note. Marit Health (2026), already listed in References, supports only the benchmark check in F8. F9-F11 are author-proposed planning assumptions. The accompanying pilot_feasibility.py records the sample approximation, cluster rounding, and illustrative budget. Appendices A-E and the compensation results are unchanged. This follow-up does not add the proposed goals-and-reversibility Appendix G.